

Human-In-The-Loop Machine Learning (HITL ML): Active Learning

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Agenda

- Annotating unlabeled data
- Sampling unlabeled data items (active learning)
- Incorporating human–computer interaction into annotation
- Implementing transfer learning

Why HITL ML?

- ~90% of ML applications are powered by supervised learning.
- Supervised models require labeled data — often created by humans.
 - Humans (often non-coders) provide these examples (training data).
- Today's AI relies heavily on intensive human feedback.
- Use cases:
 - **Autonomous Vehicles:** Thousands of hours of manual annotation (e.g., pedestrians, traffic signs).
 - **Machine Translation:** Models trained on human-translated texts for linguistic quality.
 - **Interactive AI Systems:** Systems learn more from nontechnical users than from programmers.

How should humans & ML algorithms interact?

Defining HITL ML

A set of strategies for combining human and machine intelligence.

Objectives

- Increase ML model accuracy.
- Reach target/acceptable accuracy faster
- Combine human/machine intelligence to maximize accuracy (synergy)
- Assist human tasks with ML to increase efficiency

Components of HITL ML

Annotation

- Process of enrich raw data with tags (human knowledge).
- **Purpose:** Translates expert knowledge into usable training examples for ML models.
- **Value:** The foundation of model performance — garbage in, garbage out.

Active Learning

- A strategy where the model selectively queries humans to label the most informative data points.
- **Purpose:** Maximize learning while minimizing labeling cost and time.
- **Value:** Efficient use of human effort — focusing on what matters most.

Transfer Learning

- Leveraging pre-trained models and adapting them to new, domain-specific tasks.
- **Purpose:** Reduce training time and labeled data needs, especially in data-scarce environments.
- **Value:** Get a head start and avoid cold start.

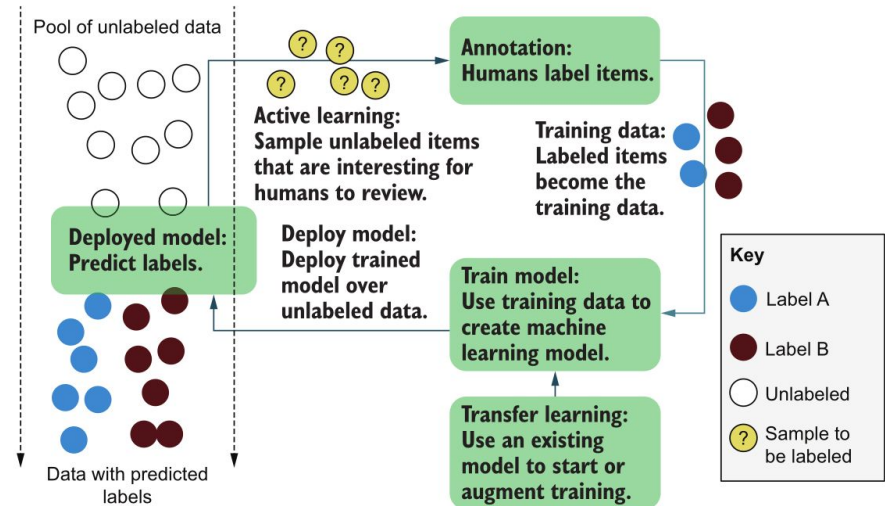
The HITL ML Process Cycle (Iterative Loop)

Start:

- **Unlabeled Data Pool**
 - Large collection of raw, unlabeled examples.

Optional Kick-Start:

- **Transfer Learning**
 - Use pretrained models to initialize learning or improve sample selection.



The HITL ML Process Cycle (Iterative Loop)

Active Learning

- Select the most "informative" or uncertain examples from the unlabeled pool.

Human Annotation

- Domain experts label selected data.

Training Dataset Creation

- Build or augment the labeled dataset.

Model Training

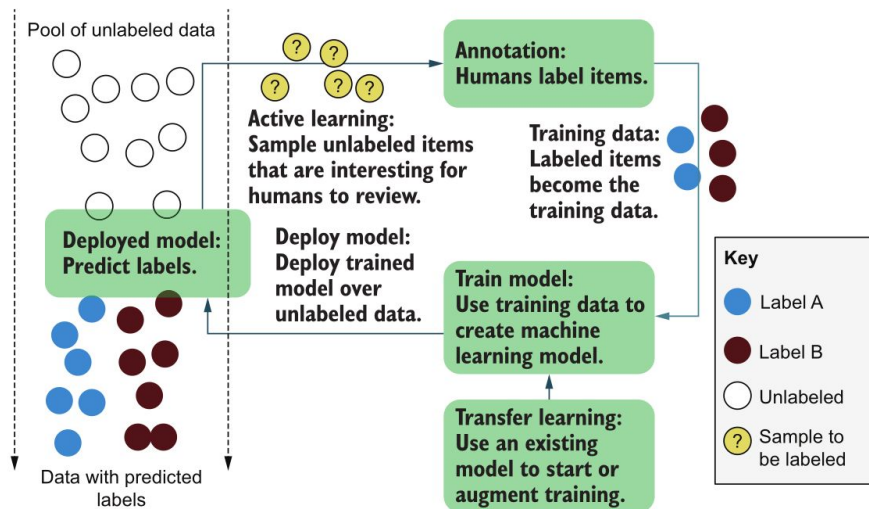
- Train or fine-tune the model on the updated labeled data.

Model Deployment

- Apply the trained model to new, unseen data.

Prediction

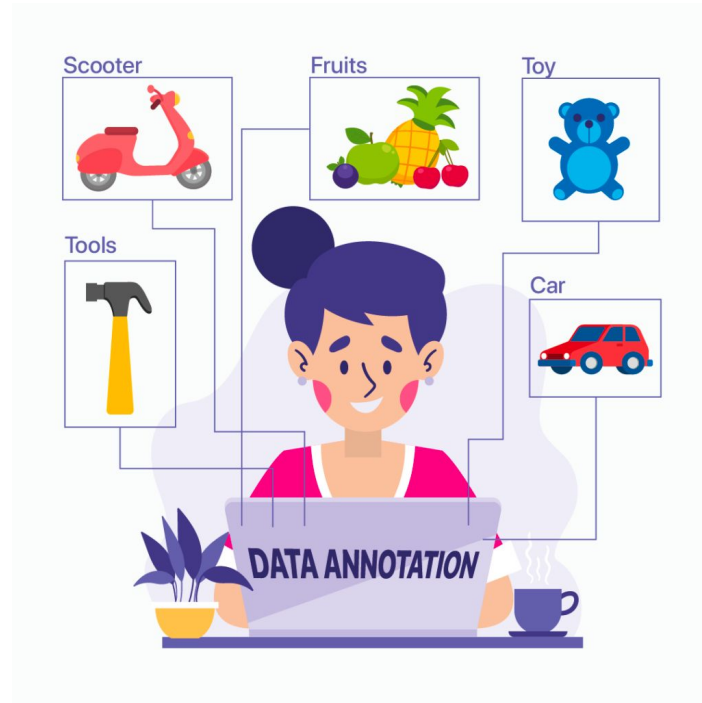
- Generate labels for new data instances.



Annotation

Annotation is the process of labeling raw data (images, text, audio, video, etc.) to transform it into structured training.

- It gives **meaning** to raw data, enabling models to learn patterns.
 - High-quality annotations are **critical for model performance**.
 - It defines what the model will learn
-
- Often the **most labor-intensive** phase in the ML pipeline.
 - Requires **precision**, consistency, and sometimes **iterative revisions**.
 - May involve **millions of labeled examples** in large-scale projects.



Annotation: Spectrum of Annotation Complexity

Simple Annotation Tasks: Tasks that can be done quickly with basic tools and minimal training.

- **Sentiment labeling** using HTML forms or spreadsheets.
(E.g., Tagging a product review as **positive**, **negative**, or **neutral**.)
- **Tools:** Web forms, Google Sheets, Label Studio
- **Characteristics:** High throughput, Suitable for crowdsourcing (e.g., Mechanical Turk), Low barrier to entry

The burgers here are **amazing.** POSITIVE

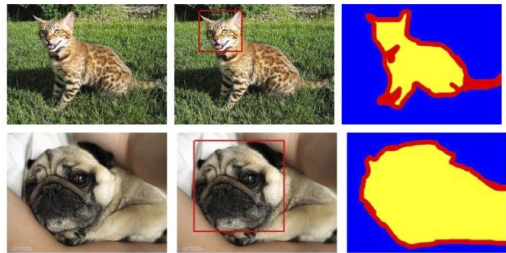
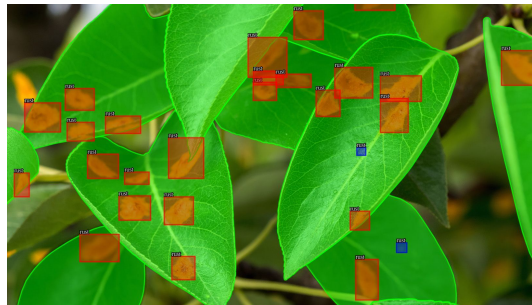
However, the appetizers were **just okay,** NEUTRAL

and the service was **terrible.** NEGATIVE

Annotation: Spectrum of Annotation Complexity

Complex Annotation Tasks: asks requiring advanced tools, domain expertise, and significant preparation.

- **Drawing bounding boxes** around moving objects in **video streams** using specialized graphical interfaces.
- **Tools:** CVAT, Labelbox, custom-built annotation platforms.
- **Challenges:**
 - Annotation fatigue and human error
 - Requires **custom engineering** (UI, storage, QA pipelines)
- **Effort:**
 - **Months of development** to create efficient workflows
 - Involves **UI/UX design**, **tool optimization**, and **QA automation**



Annotation: Data-Centric AI Shift & Academia-Industry Gap

Performance gains often come from improving data quality, not just tweaking models.

Academic ML:

- Fixed datasets (e.g., ImageNet)
- Focus on algorithmic innovation
- Ignores real-world data challenges

Industry Reality:

- Data is noisy, incomplete, ever-changing
- Success = continuous **data refinement**
- Co-optimization of **models + annotation strategies**

**In modern ML, your data is as important as — if not more than —
your model**

Annotation: Why High-Quality Annotation is Hard

- **Human Error:**
 - Critical in high-stakes applications (e.g., **autonomous vehicles**).
 - Errors are **not random noise**—they introduce **systematic bias**.
 - ***"No algorithm survives truly bad training data."***
- **Label Aggregation Challenges:**
 - **Binary/objective tasks** (e.g., spam detection) are easier to aggregate.
 - **Subjective/continuous tasks** (e.g., bounding boxes, severity ratings) are more nuanced.
 - **ML tools can assist** in reconciling disagreements.
- **Workforce Considerations:**
 - Quality depends on finding, teaching, and managing **skilled annotators**.
 - Example: Training annotators to do complex medical annotation

“Better data beats more data — and humans are essential to making it better.”

Active Learning

Active Learning is a machine learning approach where the model **selectively chooses which data samples should be labeled by humans**, rather than labeling everything.

Goal:

- **Maximize model performance** with **minimal labeled data**
- Reduce **cost**, **time**, and **annotation effort**

Why ?:

- More labeled data generally improves model performance
- But human annotation is **slow**, **costly**, and **error-prone**
- Active learning helps focus effort on the **most informative examples**

Active Learning: Three strategies

Uncertainty Sampling (Exploitation)

- Identify items near the decision boundary (model is "confused").
- Likely to be misclassified, so labeling them moves the boundary.
- **Risk: Might focus on one part of the decision boundary.**

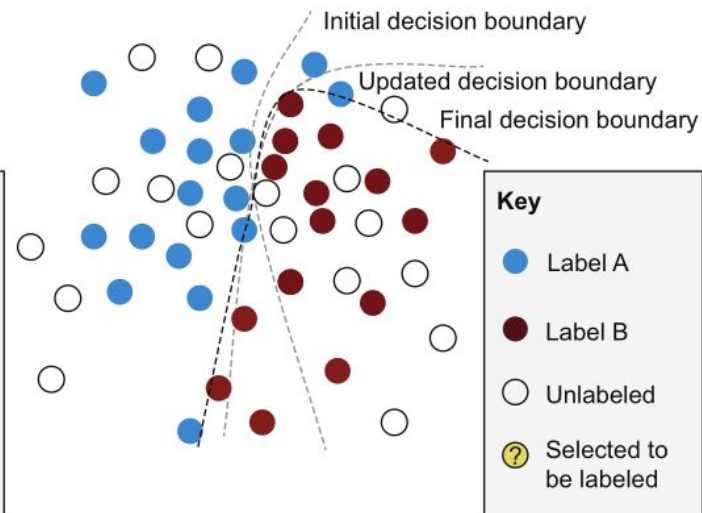
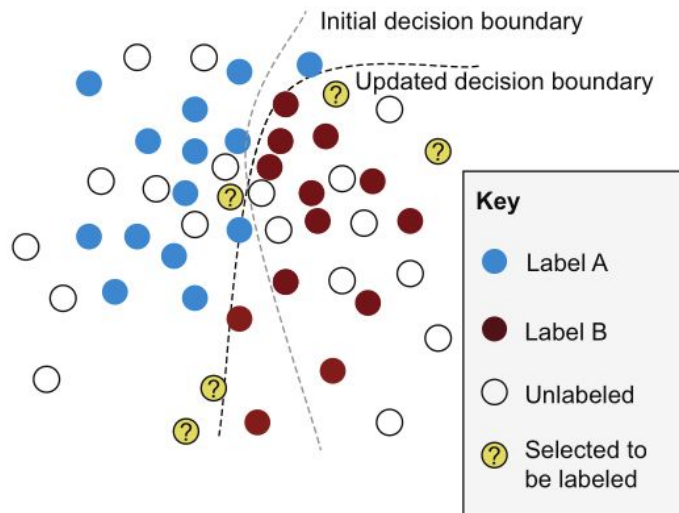
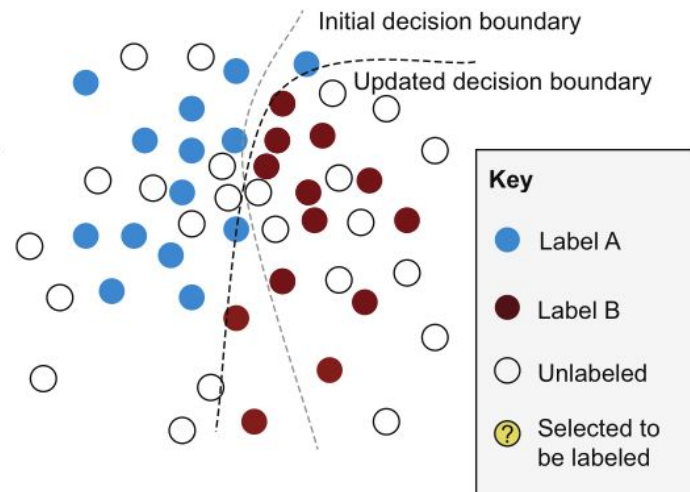
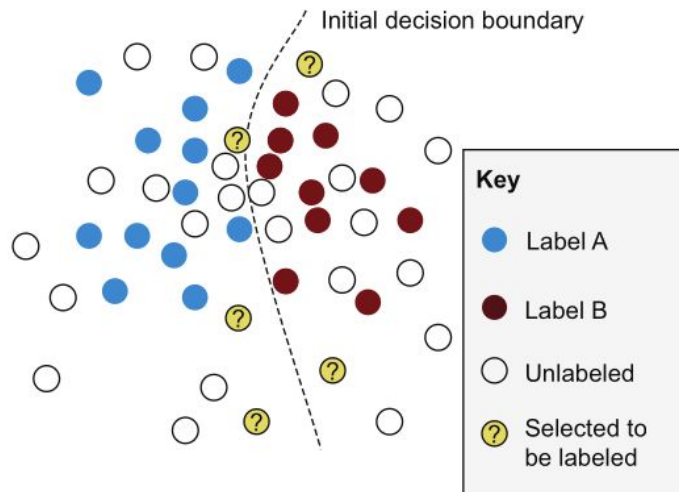
Diversity Sampling (Exploration)

- Identify underrepresented or unknown items.
- Targets new, unusual, or underrepresented items for a complete picture.
- **Risk: Might focus on outliers far from the boundary, having little impact.**

Random Sampling

- Seems simple but tricky (prefiltered data, changing data).
- Essential for gauging model accuracy and as a baseline.

Combining Strategies: Often the best approach to maximize both uncertainty and diversity.



Active Learning: Defining "Random" for Evaluation Data

Random $\neq$ Representative

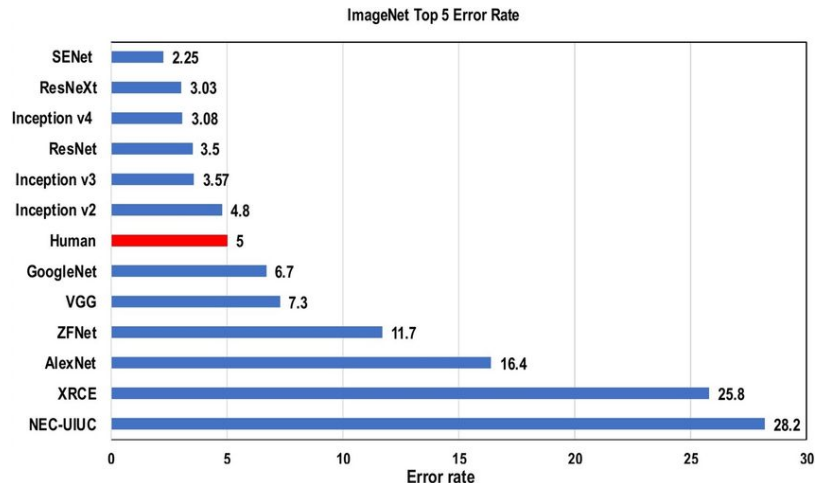
- Truly random sampling is difficult in practice: **Prefiltering bias**, **Data drift** over time or source
- Real-world data often diverges from training sets

ImageNet Case Study:

- Models achieve **high accuracy** on benchmark (ImageNet)
- But fail on **real-world images** (e.g., social media posts)
- Shows the risk of overfitting to **static, curated datasets**

A Better Approach:

- Build **representative evaluation sets**:
 - Stratify by **label distribution**, **time**, **data source**
- Create **multiple test sets**:
 - **In-domain** (e.g., curated news)
 - **Out-of-domain** (e.g., noisy social media, low-resource regions)



Active Learning: When to Use Active Learning

Use Active Learning When:

- **You can only annotate a small fraction** of your total dataset
 - Full labeling is infeasible (cost/time/resource constraints)
- **Random sampling fails to capture data diversity**
 - Leads to underrepresentation of rare or edge cases
- **Annotation costs are extremely high**
 - *Example:* Video annotation
 - **36,000 bounding boxes per minute of video**
 - **U.S. driving: ~60 trillion hours/year**
 - **Manually labeling even a fraction is unsustainable**
- **You expect data drift or edge cases**
 - Active learning helps models adapt to evolving environments

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Don't Use Active Learning When:

- **Critical require full annotation**
 - E.g., regulated medical records, financial documents
- **You operate in a highly stable, static environment**
 - Model already performs well and no new data domains are expected
- **The cost of new data exceeds the value of model improvement**
 - Important in scenarios with small performance gains or fixed ROI

Machine Learning and Human-Computer Interaction (HCI)

Machine translation

- Initial efforts to improve human translation with machine translation faced significant hurdles.
- Simply having humans correct machine translation output proved inefficient and prone to error perpetuation.
- The core challenge wasn't just the accuracy of machine translation algorithms.

The Breakthrough: A User Interface Innovation

- The real solution emerged from advancements in the user interface.
- Modern translation systems adopted **predictive text**, similar to smartphones and email.
- Translators type naturally, accepting correct machine-generated word predictions with a simple key press.
- This significantly increased translation speed when predictions were accurate.

The most significant breakthrough was in human-computer interaction (HCI), not the underlying machine learning algorithm itself.

Effective integration of human input became the crucial factor.

User interfaces: How do you create training data?

Why They Work:

- Leverage familiar **HTML form conventions**
- Follow well-established **Human-Computer Interaction (HCI)** principles
- Require **no user training** — users intuitively understand how to interact

Design Matters:

- Consistency with standard form layouts ensures **efficiency and reliability**
- Even **small deviations** (e.g., dynamic helper text) can cause confusion
- Annotation tools should **minimize friction**, not add features that surprise users

The Power of Simplicity: Binary Responses

Binary decisions (yes/no, correct/incorrect)

- Simple is often the best for quality control.
- Breaking down complex tasks into binary decisions simplifies interface design.
- This also facilitates the implementation of quality control measures.

Complex Interfaces, Complex Expectations

- More complicated annotation tasks (e.g., polygon annotation) $\Rightarrow$ more complex interface requirements.
- Users bring expectations from familiar software (e.g., Photoshop for image editing).
- Meeting these expectations can be development-intensive

The Power of Simplicity: Binary Responses

The Inefficiency of Mouse Usage

- For repetitive annotation tasks, mouse movement is inefficient.
- Aim to make the entire annotation process **keyboard-driven**.
- This includes annotation itself, form submission, and navigation.
- **Keyboard-centric workflows can significantly improve annotator rhythm.**

When a Mouse is Necessary: Rich Annotations

- If mouse usage is unavoidable, ensure the resulting annotations are rich and valuable.
- This justifies the slower input method.

Human Attention & Priming in Annotation

Why Attention Matters

- Human annotator focus and attention span are crucial for accurate training data
- Contextual effects can alter perception and introduce errors.
- ***Linguistics research demonstrates that the presence of a kiwi or kangaroo influenced accent perception.***

Priming: How Sequence Shapes Perception?

- Priming describes how context or sequence influences human perception.
- A sequence of similar tasks can bias annotator responses.
- ***For example, annotating 99 negative sentiment posts may lead to errors on the 100th post***

Diversify examples to keep the annotator focused

Basic principles for designing annotation interfaces

Favor Binary Choices

- Simplifies decision-making and UI design
- Enables easier **quality control** and consistency
- Break complex tasks into a series of **yes/no**

Diversify Task Sequences

- Prevents **priming effects** and cognitive bias
- Mix samples to keep annotators engaged and alerted
- Randomize order to reduce repetition-induced errors

Follow Established Interaction Conventions

- Build on **standard HTML forms** and familiar UI elements
- Minimize need for training or onboarding
- Avoid “clever” design that violates user expectations

Support Keyboard Shortcuts

- Improves speed, ergonomics, and flow for power users
- Reduces cognitive load compared to mouse-only interfaces

Machine-learning-assisted humans VS. human-assisted machine learning

Improve ML Models with Human Input

- Humans provide **labels**, **corrections**, or **feedback**
- Helps models learn faster, generalize better, and reduce bias

Examples

- Labeling rare edge cases
- Reviewing model predictions in active learning
- Enhancing LLMs with Human Feedback (RLHF)

Enhance Human Tasks with ML Assistance

- ML helps **humans make better decisions**
- Tasks become **faster**, **more scalable**, and **less error-prone**

Examples

- Auto-suggest in labeling tools
- Pre-annotations with confidence scores
- Autocomplete in message writing

Search Engines: A Combined Approach

Human-Assisted Machine Learning

Users train the system

- Every **click, scroll, or dwell time** is feedback.
- Clicking the 4th result over the 1st?
→ Signals “**This result is more relevant.**”
- **Patterns of behavior** become training data.
- Search ranking **adapts based on user interactions**

No need for explicit annotation — behavior becomes feedback

Machine-Learning-Assisted Humans

The system supports the user

- Search engines use **ML to rank, filter, and personalize results.**
- Algorithms trained on past behavior predict what’s most useful.
- Users get **faster, more accurate answers** with less effort.

The system evolves to better serve human intent over time.

Transfer learning to kick-start your models

Use What Already Exists

- **Existing datasets** often align closely with your goals
- Example:
 - Adapt a **product review sentiment model** for **movie reviews**
 - No need to collect labels from zero

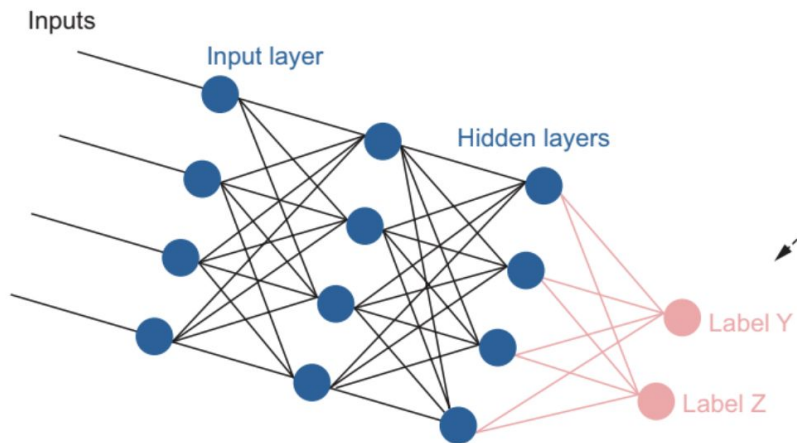
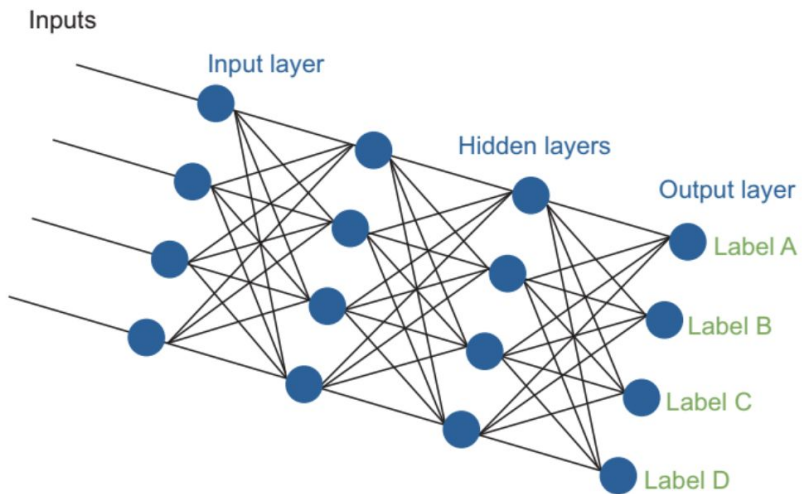
What is Transfer Learning?

- Reusing a model **trained on one task** to improve performance on a **related task**
- Saves time, data, and compute

Examples

- Use **general pre-trained models** (e.g., ImageNet, BERT, CLIP, GPT)
- **Fine-tune** them for your specific domain or use case

Most modern ML projects succeed by adapting, not reinventing.



Other labeled data



New output layer
trained to predict
the new labels

Transfer learning in computer vision

Transfer learning has significantly advanced computer vision performance.

ImageNet

- Common strategy: **Pre-train on ImageNet**, a massive dataset.
- ImageNet includes **millions of images** across **1,000 diverse categories** (e.g., *animals, objects, sports, birds, etc.*).

Reusing Learned Features

- Models trained on ImageNet learn **general visual features** like:
 - **Edges, textures, shapes**
- These features are **transferable to other tasks**
 - Examples: medical imaging, object detection).

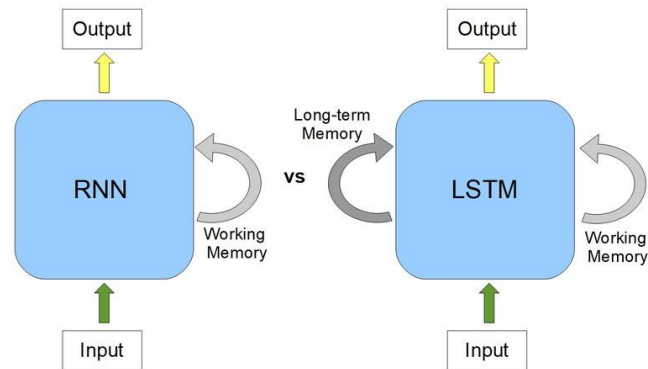


Retraining the last layers of ImageNet models enables accurate classification of new tasks, even with limited data.

Transfer Learning in Natural Language Processing: Early Struggles

- **LSTMs & RNNs** dominated early NLP.
- **Problems:**
 - Poor generalization across tasks.
 - Struggled with long-range dependencies.
 - Required large amounts of task-specific labeled data.

*Progress in **text transfer learning** was slow compared to vision.*

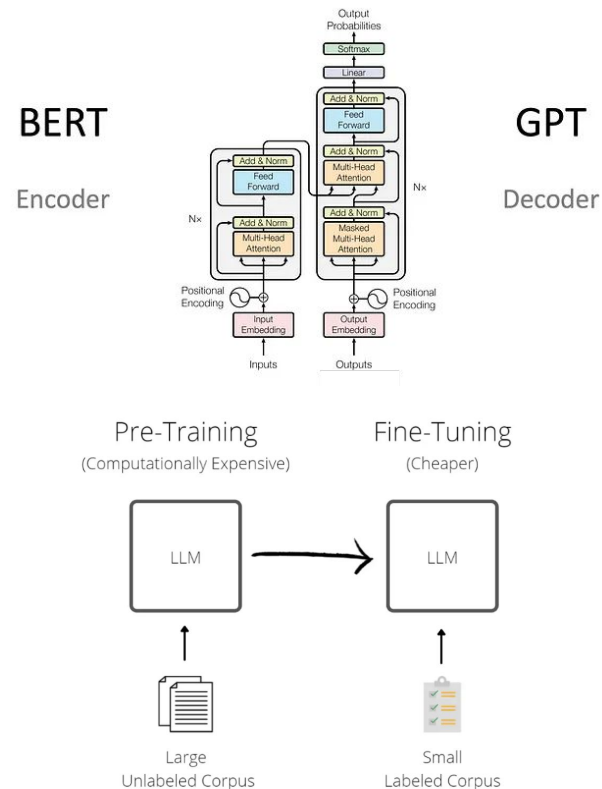


Transfer Learning in Natural Language Processing: Transformer and LLM Breakthrough

- **Transformers (Vaswani et al., 2017):**
 - Used **self-attention** for better context handling.
 - Enabled parallel training over large corpora.
- **BERT, GPT:**
 - Pre-trained on massive text (Wikipedia, web, books).
 - Learned general-purpose language representations.

Marked a shift: Pre-train once, fine-tune many tasks.

Transfer learning & LLMs significantly improved efficiency and scalability in NLP applications.



Learning Word Meaning from Context

“You shall know a word by the company it keeps.” – Firth, 1957

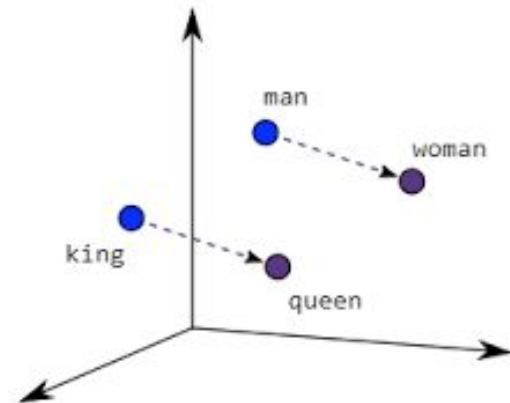
- Words used in similar contexts tend to have similar meanings.
 - Example: *doctor* and *surgeon* often appear in medical contexts.

How Models Learn Meaning

- Analyze word usage across **thousands of contexts**.
- Represent words as **vectors** capturing contextual patterns.
 - Similar contexts → closer vectors.

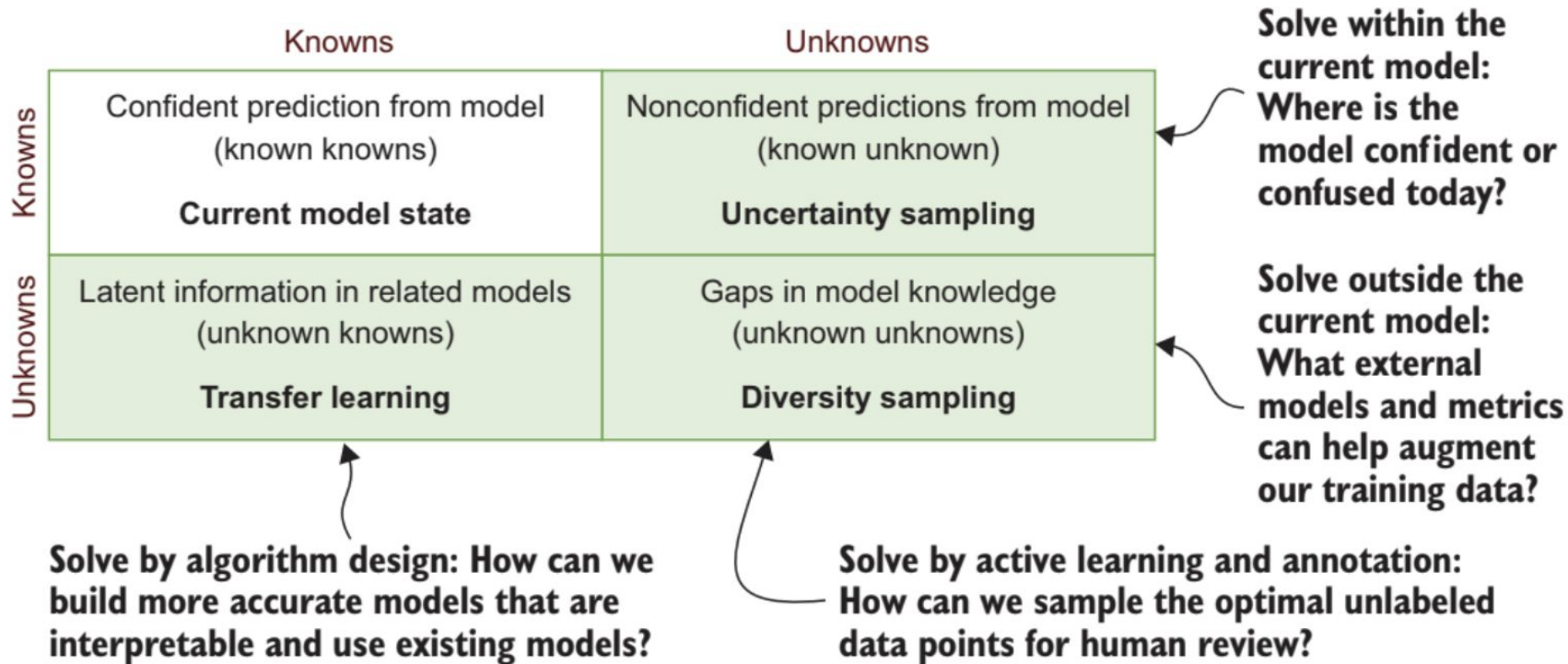
Free Supervision via Context Prediction

- Self supervised learning: Predict missing words from surrounding context.
 - Example: “My ___ is cute. He ___ playing.” ⇒ “dog”, “is”
- Enables **training on massive unlabeled corpora**, powering transfer learning.



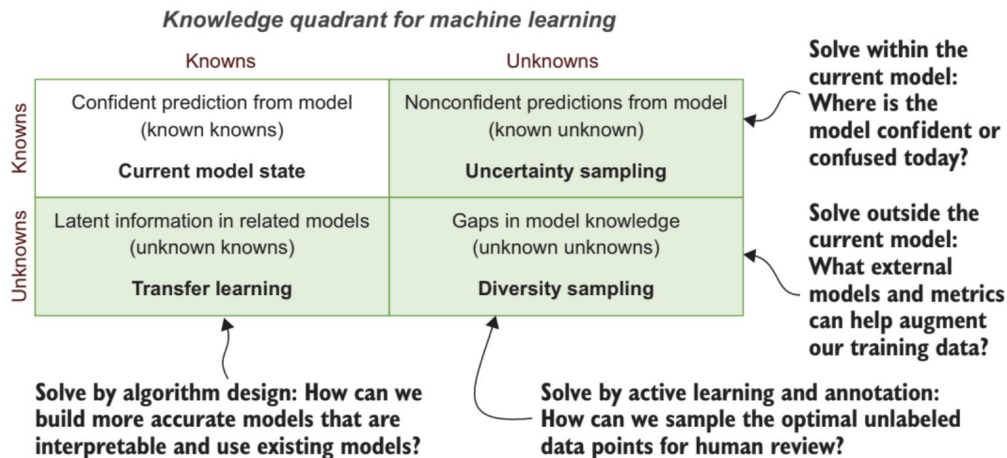
Context prediction is the engine behind LLMs' ability to learn and transfer word meaning across tasks.

Knowledge quadrant for machine learning

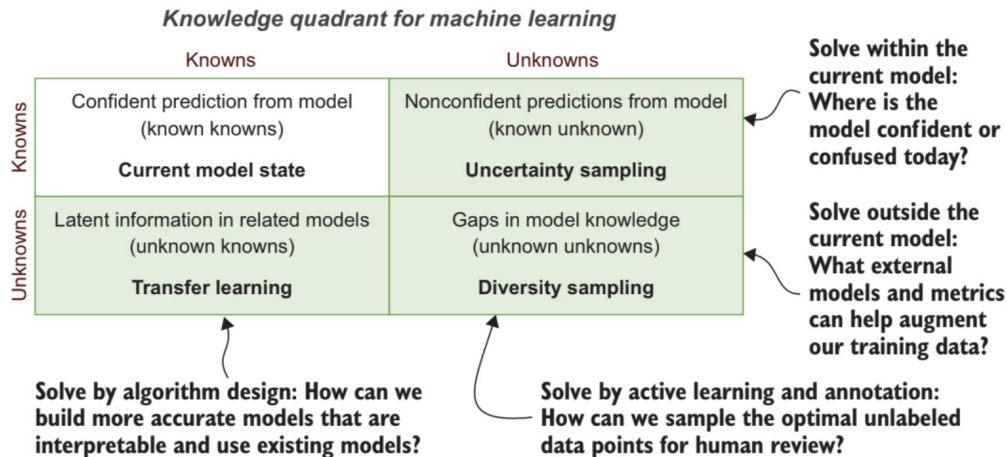


The four quadrants are

- **Known knowns**—What your machine learning model can confidently and accurately do today. This quadrant is your model in its current state.
- **Known unknowns**—What your machine learning model cannot confidently do today. You can apply uncertainty sampling to these items.
- **Unknown knowns**—Knowledge within pretrained models that can be adapted to your task. Transfer learning allows you to use this knowledge.
- **Unknown unknowns**—Gaps in your machine learning model. You can apply diversity sampling to these items



- The top row captures your **model's knowledge**.
- The bottom row captures **knowledge outside your model**
- The left column can be **addressed by the right algorithms**
- The right column can be **addressed by human interaction**



Key Takeaways

- **HITL ML is an Iterative Process**
 - Combines human judgment with machine learning to continuously improve models.
- **Annotation Techniques Are Critical in ML**
 - Essential for creating high-quality training data efficiently and accurately.
- **Human-Computer Interaction (HCI) Enhances the Annotation Process**
 - Enables the design of effective, user-friendly annotation tools.
- **Active Learning Optimizes Label Acquisition**
 - *Uncertainty sampling*: Targets data the model is least confident about.
 - *Diversity sampling*: Selects a broad range of examples to maximize learning.
- **Transfer Learning Accelerates ML Applications**
 - Leverages pre-trained models to reduce annotation effort and improve accuracy.

Reference book

